

Jeevesh Singale

AI Engineer | LLMs & Agentic Solutions | Python
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Professional Summary

AI Engineer with 3.5 years of experience taking LLM and agentic solutions through the full lifecycle, research and prototyping through to deployment and monitoring. My clinical RAG agent runs a LangGraph agent with tool-use (clinical calculators), rule-based care-gap detection across synthetic patient cases, and a RAGAS-compatible evaluation harness with CI-enforced quality gates, deployed live with 153 passing tests. A second agentic platform, ExpertIQ Copilot, applies the same end-to-end approach with vector and knowledge-graph retrieval. My deep learning exposure centres on LSTM sequence modelling (TensorFlow/Keras) from my MSc dissertation, one area in depth rather than broad claims across the field. Hands-on with Databricks and Apache Spark for data pipelines on AWS and Azure. Have mentored junior team members during an onsite apprenticeship in Pune. Anthropic MCP certified. MSc Information Systems & Operations Management, University of Nottingham.

Technical Skills

LLMs & Agentic Frameworks: LangGraph, LangChain, Agentic Workflows (Multi-Step, Tool-Use), Prompt Engineering, Structured Output Design

RAG Patterns & Retrieval: Pinecone, ChromaDB, Hybrid Dense/Sparse Retrieval (BM25), Cohere Embeddings & Rerank, Chunking, HyDE

Evaluation & Quality Gates: RAGAS-compatible harnesses, CI-enforced quality thresholds, LangSmith (custom evaluators), Pytest (153-test suite), SHAP, LIME

Applied ML: Python, XGBoost, Random Forest, LSTM (TensorFlow/Keras), scikit-learn

Security & Governance: JWT Authentication, SSRF Prevention, Prompt-Injection Defense, Credential Protection, Ingestion Audit Trails, Safety Guardrails

Data Engineering: Databricks, Apache Spark, SQL, PostgreSQL, dbt, Apache Airflow, ETL/ELT

Programming & APIs: Python (primary, production-grade), TypeScript, FastAPI, REST API Design, Microservices

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions CI/CD, Vercel

Professional Experience

AI-First Software Engineer & Data Analyst | Risidio, London, UK *Sep 2025 – Present*

- Shipped a Web3 revenue dashboard (Next.js/TypeScript) integrating ERC-4337 smart contract logic for automated multi-party royalty distribution, achieving sub-4-second on-chain settlement latency and cutting finance reporting time in half.
- Consolidated multi-source financial data into automated ETL pipelines, then hardened the services for production with JWT authentication, bcrypt hashing, and rate limiting.

Web Data Analyst & Developer | Javish Jewellery, Mumbai, India *Apr 2025 – Sep 2025*

- Built multi-source ETL/ELT pipelines on **Databricks** and Apache Spark that cut manual reporting time in half and gave senior leadership real-time input for sales strategy decisions.
- Designed Power BI and Tableau dashboards around an A/B-tested KPI framework, which sped up leadership decision-making by 20%.

Data Science Apprentice | Global EduTech, Pune, India *Jun 2022 – Mar 2025*

- Ran ETL workflows across AWS and Azure at 99%+ uptime and deployed 5+ ML models with full MLOps observability. Every model shipped with SHAP/LIME explainability and documentation stakeholders could actually use.
- For my MSc dissertation I compared Random Forest, XGBoost, and LSTM (TensorFlow/Keras) on 1M+ telecom customer records. Random Forest won out at 96% accuracy, and the resulting retention strategy projected \$2–3M in annual savings.
- Mentored incoming data science apprentices for 3 months during an onsite placement in Pune, pairing on model-building fundamentals and ETL practices.

Key Projects

Clinical Evidence RAG Agent | Personal Project · Live Deployment

Jun 2026 – Present

- Built a LangGraph agent (FastAPI backend) that classifies each query into one of six intents — guideline, workflow, calculator, unsafe-request, insufficient-evidence, or out-of-domain — then routes it accordingly via tool-use: clinical calculators (BMI, MAP, pulse pressure, eGFR) and a rule-based care-gap checker that flags missing follow-ups across 5 synthetic patient cases without relying on an LLM for the underlying logic. Deployed live, 153 tests passing across 15 test areas.
- Layered a curated knowledge graph (28 wikilinked concept files) in front of hybrid dense/sparse retrieval (Cohere embeddings and reranking): canonical facts are checked first, RAG is the fallback, not the default.
- Built a RAGAS-compatible evaluation harness with CI-enforced quality thresholds and an ingestion pipeline that writes a SHA-256-hashed audit manifest per source, so every answer traces back to a specific, verifiable document version.
- Hardened the whole path for production: a safety classifier refuses unsafe medical-advice requests before any retrieval or LLM call, credential fields are excluded from logs, SSRF checks block private/reserved IPs on PDF ingestion, and the container runs as a non-root user.

ExpertIQ Copilot: AI Expert Discovery Platform | Personal Project · Live Demo

Apr 2026 – Jun 2026

- Architected a 6-node LangGraph workflow combining ChromaDB vector search with NetworkX knowledge-graph traversal for retrieval, enabling hybrid retrieval that surfaces both semantically and relationally relevant expert profiles across a curated domain corpus.
- Implemented continuous evaluation through LangSmith against a 47-case regression suite, enforcing retrieval and response-quality gates on every commit rather than spot-checked by hand.
- Designed structured output schemas for expert ranking with Cohere reranking, producing ranked candidate lists with confidence scores and evidence trails for each recommendation.
- Deployed as a containerised FastAPI live demo with sub-second query latency across multi-hop graph traversals and vector similarity searches.

Education & Certifications

MSc Information Systems & Operations Management, University of Nottingham, UK

Sep 2023 – Sep 2024

BEng Information Technology, Mumbai University, India

Aug 2019 – Jun 2023, GPA 9.2/10

Certifications:

- Associate AI Engineer for Developers, DataCamp, 2026
- RAG Bootcamp, KodeKloud, 2026
- Anthropic MCP & Advanced Topics, 2026
- Databricks Generative AI Fundamentals, 2025